

SHAURYA TIWARI

AI/ML Systems & Full-Stack Development

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PROFILE

Application engineer who builds end-to-end business systems — from requirements and architecture through web interfaces, APIs, databases, third-party and cloud integrations, automated testing, deployment, and operational reliability. Built a 13-stage document intelligence platform with 64 deterministic tests and re-engineered a procurement workflow for 3x faster execution.

TECHNICAL SKILLS

Languages & Fundamentals: Python, C++, SQL, Data Structures & Algorithms, Object-Oriented Programming, Git

Application Engineering: Requirements analysis, system design & architecture, SDLC, web application development, REST APIs, third-party integration, automated testing, debugging, documentation, deployment, support & maintenance

Web, Backend & Data: Streamlit, Flask, LangChain, SQLite, ChromaDB, BM25, vector search, Pandas, NumPy, data ingestion, ETL & data pipelines

Cloud & AI Integrations: Azure AI Inference SDK, Azure AI Foundry-compatible APIs, GitHub Models, Groq API, Serper API, Ollama, PyTorch, scikit-learn, sentence-transformers, RAGAS

ENGINEERING PROJECTS

Multi-Format Document Intelligence Platform

Python · ChromaDB · BM25 · Azure AI SDK · Hybrid Retrieval · Automated Testing

- Architected and built a 13-stage ingestion-to-evaluation pipeline for 11 file types across text, spreadsheets, Office documents, PDFs, and images, with OCR/vision fallback, token-aware chunking, persistent indexing, summarization, and evaluation.
- Designed document routing and hybrid retrieval using ChromaDB, BM25, reciprocal-rank fusion, and cross-encoder reranking; added interchangeable adapters for GitHub Models, Azure AI Foundry, and Ollama without changing agent logic.
- Productionized the system with per-file isolation, corruption/password/zip-bomb checks, stale-index cleanup, bounded retries, persistent memory, vision fallback, and prompt-injection detection; wrote 64 deterministic offline tests across six subsystems.

B2B Procurement Workflow Platform (Third Eye)

Python · LangChain · Llama-3 70B · Streamlit · REST APIs · SQLite

- Built an end-to-end business process application integrating a Streamlit interface, LangChain orchestration, Serper and Groq APIs, web scraping, and SQLite persistence to discover tenders, parse documents, and generate tailored proposals.
- Re-engineered unstable orchestration into sequential chains, improving execution speed 3x and eliminating observed crashes; separated scraping from the UI through a SQLite-backed asynchronous workflow.
- Implemented deterministic domain validation that rejects irrelevant signals before proposal generation, reducing hallucinated and unusable outputs.

RESEARCH EXPERIENCE

Machine Learning Research — M.Tech Thesis, IIT Roorkee

Jan 2025 – May 2026

- Built a six-channel segmentation pipeline from BSE microscopy and five EDS elemental maps; achieved 0.94+ mIoU, 95%+ pixel accuracy, and F1 > 0.90 across six classes on unseen cross-sample data.
- Benchmarked Mask R-CNN against U-Net++ and selected a ResNet34-backed U-Net++, improving mIoU by 6+ points over the prior published domain benchmark.
- Developed memory-safe sliding-window inference for high-resolution images on 16 GB VRAM and automated post-processing for 18,996 detected grains, with no boundary artifacts or out-of-memory failures.

EDUCATION

M.Tech, Space Science & Technology — Indian Institute of Technology Roorkee

2024 – 2026

CGPA: 8.80 / 10

M.Sc, Applied Geology & Geoinformatics — National Institute of Technology Durgapur

2022 – 2024

CGPA: 8.14 / 10

B.Sc (Hons), Geology — Banaras Hindu University

2018 – 2021

CGPA: 8.30 / 10

LEADERSHIP & RECOGNITION

Placement Coordinator, IIT Roorkee (2024–2026): Coordinated placement drives with 50+ companies and managed recruiter communication, scheduling, and student–industry stakeholder coordination.

Codeforces Expert (1610 rating): Ranks in the top 14% of active competitive programmers on Codeforces, with a best global contest rank of 797.